

Irrigation Pump Automation and Remote Monitoring System

Custom PCB Design for Industrial Field Monitoring

Project Overview

This project focused on the design and implementation of a custom electronic monitoring system for irrigation pumps used in large sugarcane fields. These pumps were in remote agricultural areas, making manual supervision difficult and inefficient. The system was developed to monitor pump operation, fuel usage, water flow, and fuel level from a centralized location.

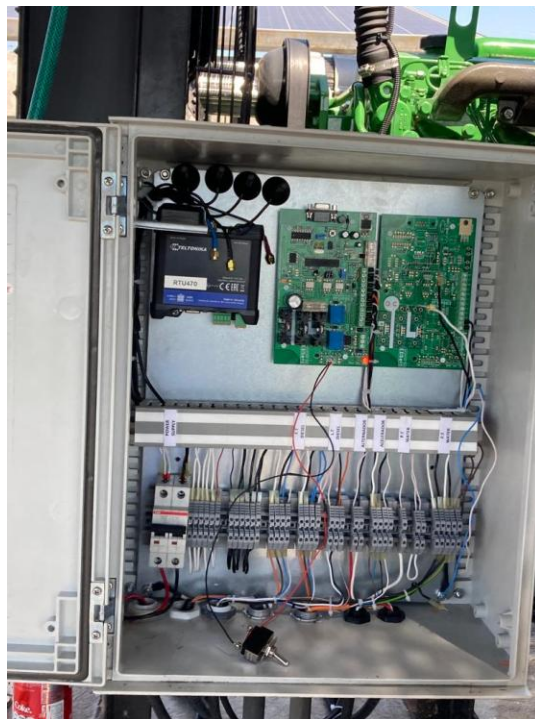


Figure 1. Control enclosure with custom PCB and communication module.

Motivation

Sugarcane irrigation systems often operate across large agricultural fields located far from the main facilities. Because of the distance, it was difficult to know when each pump was running, how much fuel was being consumed, how much water was being pumped, or whether the system was operating correctly.

System Concept

The system was based on a custom PCB integrated with sensors, industrial wiring, protection elements, and a communication module. The PCB collected information from the irrigation pump system, including engine status, fuel consumption, fuel level, and water flow. This information could then be transmitted remotely for monitoring and analysis.



Figure 2. Field installation and testing near the irrigation pump.

My Role

My contributions to this project included:

- Designed custom PCBs for remote monitoring of irrigation pump systems.
- Selected electronic components, sensors, connectors, and interface circuits.
- Supported PCB manufacturing, assembly, and verification.
- Integrated the PCB with field sensors, wiring terminals, and communication hardware.

- Tested the system directly in the field with operational irrigation pumps.
- Assisted with troubleshooting, installation, and validation under real industrial conditions.
- Collaborated with technicians and field personnel during deployment and testing.

Methodology

The project began by identifying the variables that needed to be monitored in each irrigation pump: fuel usage, water flow, engine on/off state, and fuel level. Based on these requirements, the electronic system architecture was designed, including sensor inputs, power connections, signal conditioning, communication hardware, and field wiring interfaces.



Figure 3. Complete pump station with monitoring electronics installed.

Key Technical Areas

This project demonstrates experience in:

- PCB design and validation
- Industrial automation
- Remote monitoring systems
- Communication module integration
- Troubleshooting and field deployment
- Agricultural automation systems

Results and Demonstrations

The project resulted in a custom PCB-based monitoring system installed in irrigation pump stations. The system enabled the collection of operational data from remote pumps and supported improved monitoring of fuel consumption, pump activity, water flow, and fuel level.

Industrial Relevance

This project is relevant to industrial automation, agricultural technology, and remote equipment monitoring. It demonstrates how custom electronics can be used to transform isolated mechanical systems into monitored industrial assets.

Project Status

This project was developed during my work at Ingenio La Unión. My contribution focused on PCB design, component selection, hardware verification, field integration, and testing of the remote monitoring system for irrigation pumps. The project demonstrated the use of custom electronics to improve monitoring and operational control of agricultural pumping systems.